x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
1	1				
		7			
2	8		12		
		19		6	
3	27		18		0
		37		6	
4	64		24		0
		61		6	
5	125		30		0
		91		6	
6	216		36		0
		127		6	
7	343		42		
		169			
8	512				

$$y_{(x=x_n+nh)} = y_n + \frac{n}{1!}\nabla y_n + \frac{n(n+1)}{2!}\nabla^2 y_n + \frac{n(n+1)(n+2)}{3!}\nabla^3 y_n + \ldots$$

To find y at $x = 7.5 \therefore x_n + nh = 7.5$, $x_n = 8, h = 1 \Rightarrow n = -0.5$

$$y_{(x=7.5)} = 512 + \frac{-0.5}{1!}169 + \frac{-0.5(-0.5+1)}{2!}42 + \frac{-0.5(-0.5+1)(-0.5+2)}{3!}6$$

$= 421.88$

Example 5.20

From the following table of half- yearly premium for policies maturing at different ages. Estimate the premium for policies maturing at the age of 63.

Age	45	50	55	60	65
Premium	114.84	96.16	83.32	74.48	68.48

Solution:

Let age = x and premium = y

To find y at $x = 63$. So apply Newton's backward interpolation formula.

$$y_{(x=x_n+nh)} = y_n + \frac{n}{1!}\nabla y_n + \frac{n(n+1)}{2!}\nabla^2 y_n + \frac{n(n+1)(n+2)}{3!}\nabla^3 y_n + \ldots$$

To find y at $x = 63 \quad \therefore x_n + nh = 63$, $x_n = 65, h = 5 \Rightarrow n = -\frac{2}{5}$

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
45	114.84				
		-18.68			
50	96.16		5.84		
		-12.84		-1.84	
55	83.32		4		0.68
		-8.84		-1.16	
60	74.48		2.84		
		-6			
65	68.48				

$$y_{(x=63)} = 68.48 + \frac{\frac{-2}{5}}{1!}(-6) + \frac{\frac{-2}{5}\left(\frac{-2}{5}+1\right)}{2!}2.84 + \frac{\frac{-2}{5}\left(\frac{-2}{5}+1\right)\left(\frac{-2}{5}+2\right)}{3!}(-1.16) + \frac{\frac{-2}{5}\left(\frac{-2}{5}+1\right)\left(\frac{-2}{5}+2\right)\left(\frac{-2}{5}+3\right)}{3!}(0.68)$$

$= 68.48 + 2.4 - 0.3408 + 0.07424 - 0 - 0.028288$

$y(63) = 70.437.$

Example 5.21

Find a polynomial of degree two which takes the values.

x	0	1	2	3	4	5	6	7
y	1	2	4	7	11	16	22	29

Solution:

We will use Newton's backward interpolation formula to find the polynomial.

$$y_{(x=x_n+nh)} = y_n + \frac{n}{1!}\nabla y_n + \frac{n(n+1)}{2!}\nabla^2 y_n + \frac{n(n+1)(n+2)}{3!}\nabla^3 y_n + \ldots$$

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